

Multimodal Fusion for Remote Sensing Land Cover Segmentation

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I. LITERATURE SURVEY

Recent studies in remote sensing segmentation show that vision-only models can struggle with complex scenes, intra-class variation, and limited semantic understanding [1]. Therefore, several studies have explored multimodal segmentation methods that use textual information to support visual feature learning. TAMS and FSVLM represent methods that generate textual descriptions and integrate them into segmentation. TAMS adopts a UNetFormer-based architecture with a BERT text encoder and aligns image and text features using cross-attention and contrastive learning [2]. FSVLM, applied to farmland segmentation, combines a segmentation model with a vision-language framework. It shows that textual descriptions can capture spatial, environmental, and temporal relationships that are difficult to model using visual features alone. It also reports that the level and content of textual descriptions affect segmentation performance [3].

CLIP2RS and RemoteCLIP follow another direction by using pretrained vision-language representations. CLIP2RS adapts CLIP for remote sensing segmentation with textual prompts. It also uses a dual-granularity alignment strategy that aligns both image-level and pixel-level features. This strategy helps address intra-class variation and class imbalance [1]. RemoteCLIP extends this idea by unifying heterogeneous annotations into image-text pairs. It then uses these pairs to train a domain-adapted vision-language foundation model [4]. Overall, these studies show that textual information can improve segmentation. However, existing studies mainly focus on multimodal alignment or adapting pretrained representations to the remote sensing domain. A research gap remains in systematically comparing the contribution of textual description variants to segmentation performance under controlled experimental conditions. This project addresses this gap by comparing textual description variants within the same model framework. It also evaluates whether image-level class distribution information can provide additional benefit to the text-augmented segmentation model when used as an auxiliary learning signal during training.

II. PROJECT PROPOSAL

To address the identified limitations, this study proposes a multimodal semantic segmentation approach that investigates whether textual descriptions can improve remote sensing segmentation performance. The primary task is semantic segmentation, where the model predicts a pixel-level land-cover map from an input image. The main research question is whether incorporating textual descriptions improves segmentation performance compared to a vision-only baseline.

To answer this question, a UNetFormer-based architecture will be used as the visual backbone. Textual features will be extracted using a pretrained vision-language model, CLIP. These textual embeddings will be integrated into the

segmentation model through text-visual cross-attention modules applied at multiple decoder stages.

Two ablation studies will also be conducted. The first will compare different textual description variants within the same model framework to evaluate their relative contribution to segmentation performance. The second will examine whether image-level class distribution information can provide additional benefit when used as a train-time auxiliary regularizer. Model performance will be assessed using pixel-level mean Intersection over Union, mIoU, as the primary metric, with per-class IoU and overall accuracy as secondary metrics.

III. PROOF OF CONCEPT

A proof of concept study evaluated whether incorporating textual descriptions improves segmentation performance. The dataset was prepared by aligning images, segmentation masks, and captions, followed by exploratory analysis of class distribution and caption characteristics. An image-only UNetFormer baseline was trained, with final evaluation performed on the held-out test set. To assess the contribution of textual information, a multimodal variant was implemented by integrating a pretrained CLIP text encoder with UNetFormer through text-visual cross-attention modules applied at multiple decoder stages. In this proof of concept, a single hybrid caption type was used as the textual input, and the text encoder was kept frozen while only the cross-attention layers were trained. The multimodal model showed consistent improvement over the image-only baseline, with approximately 9% relative gain in test mIoU. These results indicate that textual descriptions provide complementary information that supports visual feature learning in segmentation.

The following sections extend this proof of concept by evaluating all textual description variants, comparing the selected vision-only baseline with multimodal configurations, and testing the auxiliary regularizer. These experiments provide a more complete evaluation of the proposed framework. Further improvements may include data augmentation, domain-adapted vision-language models, longer training schedules, and more advanced text-vision interaction mechanisms.

IV. RESULTS

This section presents the experimental setup and the main segmentation results. The dataset contains 10,000 aerial RGB images at 256×256 resolution. Each image has a pixel-level mask for seven land-cover classes, namely Tree, Shrub, Grass, Crop, Built-up, Barren, and Water. The dataset is strongly imbalanced. Grass covers about 45% of labeled pixels, while Shrub covers only 0.8%. RGB mask colors are converted into class-index masks before training. The original class mapping is preserved during this conversion.

The dataset is split into 7,000 training, 1,500 validation, and 1,500 test images. Stratified sampling is applied based on the dominant class of each image. The same split is used in all experiments to keep the comparison fair. All models are trained for 30 epochs with AdamW, a learning rate of $6e-4$, weight decay 0.01, batch size 8, and weighted cross-entropy loss.

Two vision-only baselines from different architectural families are trained to establish the reference point. UNetFormer uses a CNN backbone with a transformer-style decoder and has about 12 million parameters. Swin-UperNet uses a transformer backbone with a Swin-Tiny encoder and a UPerNet decoder, and has about 60 million parameters. UNetFormer reaches a test mIoU of 0.649, while Swin-UperNet reaches 0.627. UNetFormer performs better on all seven classes despite being smaller, so it is selected as the primary baseline for the multimodal experiments.

The multimodal experiments use the selected UNetFormer baseline and add a frozen CLIP text encoder with gated cross-attention in the decoder. The gate parameter starts at zero and is learned during training, so textual information is introduced gradually without disturbing the pretrained visual backbone. The model is trained separately for each of the five textual description variants in the dataset. The best test mIoU is obtained with text_qwen3-4b, reaching 0.697. Vision-only textual description variants provide limited benefit. This indicates that textual information improves segmentation, but the magnitude of the improvement depends on the description type. The detailed comparison between variants is presented in the next subsection.

After identifying text_qwen3-4b as the best text-augmented configuration, a KL-based auxiliary regularizer is added on top of this model. The KL term is computed between the predicted class distribution and the image-level class distribution. It is used only during training and is not provided during inference. With a KL loss coefficient of 0.5, the final model reaches a test mIoU of 0.705.

Table I summarizes the per-class IoU and overall mIoU for three configurations, namely the vision-only baseline, the best text-augmented model, and the final model with the auxiliary regularizer.

TABLE I. PER-CLASS IOU ON THE TEST SET

Class	UNetFormer	Multimodal	Multimodal + KL
Tree	0.861	0.876	0.874
Shrub	0.248	0.326	0.352
Grass	0.763	0.806	0.813
Crop	0.742	0.816	0.818
Built-up	0.563	0.589	0.621
Barren	0.487	0.567	0.581
Water	0.878	0.899	0.874
mIoU	0.649	0.697	0.705

V. BENCHMARKING

Benchmarking first compares the vision-only UNetFormer baseline with the text-augmented model to test whether textual information improves segmentation performance. The best text-augmented configuration improves test mIoU from 0.649 to 0.697, corresponding to a 7.4% relative gain. This shows that adding textual information provides a clear improvement over the visual baseline.

The next comparison examines whether the contribution depends on the textual description variant. The text-only variant, text_qwen3-4b, gives the strongest result. The two hybrid variants, hybrid_gemma3-4b and hybrid_qwen3-vl-8b, also improve the baseline, with relative gains of 6.8% and 6.4%, respectively. In contrast, the two vision-only caption variants, vision_gemma3-4b and vision_qwen3-vl-8b, provide little or no gain, with relative changes of -1.5% and +0.5%. This suggests that the benefit does not come from adding any text input. The textual description needs to provide reliable semantic information about the land-cover classes.

The auxiliary regularizer ablation evaluates whether the KL term provides additional benefit beyond the best text-augmented configuration. Adding the KL term on top of text_qwen3-4b improves test mIoU by 1.1% relative. This brings the total improvement over the vision-only baseline to 8.6%. The modest magnitude of the gain is consistent with the role of an auxiliary regularizer. It complements the existing segmentation supervision rather than dominating the objective.

A separate methodological check tests whether the multimodal benefit comes from qualitative caption content or from numerical class proportions. Some text-rich captions explicitly include class percentages, which may raise the concern that the model benefits from numerical proportions rather than semantic content. To test this, the three text-rich caption variants are retrained after removing all numerical percentages while preserving qualitative descriptors. The resulting test mIoU values remain within $\pm 0.81\%$ of the original values. This change is small compared with the main multimodal gain, suggesting that the benefit mainly comes from qualitative semantic content rather than numerical proportions.

The per-class breakdown in Table I shows where the improvements concentrate. Compared to the vision-only baseline, the text-augmented model achieves the largest relative gains on weaker or under-represented classes. Shrub improves by 31.7%, Barren by 16.3%, and Crop by 10.0%. Adding the KL regularizer further improves several difficult classes, with the largest additional gains observed for Shrub and Built-up. Dominant classes such as Tree and Water show smaller relative changes because the baseline already performs well on them. This pattern suggests that the text branch provides complementary semantic information where visual features alone are less reliable. It also supports the interpretation that the improvements are component-driven rather than the result of simple hyperparameter tuning.

VI. REFERENCES

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